

Practice Sheet 03

Problem 1

Let $f(x, y) = x^2y + e^{3x+y}$. Find all second-order partial derivatives.

Solution. First order partial derivatives:

$$\frac{\partial f}{\partial x} = 2xy + 3e^{3x+y}, \quad \frac{\partial f}{\partial y} = x^2 + e^{3x+y}.$$

Second order:

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2} &= 2y + 9e^{3x+y}, & \frac{\partial^2 f}{\partial y^2} &= e^{3x+y}, \\ \frac{\partial^2 f}{\partial x \partial y} &= 2x + 3e^{3x+y}, & \frac{\partial^2 f}{\partial y \partial x} &= 2x + 3e^{3x+y}. \end{aligned}$$

The mixed partials are equal. □

Problem 2

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Compute $f_{xy}(0, 0)$ and $f_{yx}(0, 0)$ and show they differ.

Solution. First compute the first partials at the origin by definition:

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - 0}{h} = 0, \quad f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k) - 0}{k} = 0.$$

For $(x, y) \neq (0, 0)$ we simplify $f(x, y) = \frac{x^3y - xy^3}{x^2 + y^2}$. Differentiate:

$$f_x = \frac{(3x^2y - y^3)(x^2 + y^2) - (x^3y - xy^3)(2x)}{(x^2 + y^2)^2}.$$

To find $f_{xy}(0, 0)$, evaluate $f_x(0, y)$ for $y \neq 0$:

$$f_x(0, y) = \frac{(0 - y^3)(y^2) - 0}{(y^2)^2} = \frac{-y^5}{y^4} = -y.$$

Hence

$$f_{xy}(0, 0) = \lim_{k \rightarrow 0} \frac{f_x(0, k) - f_x(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{-k - 0}{k} = -1.$$

Now compute f_y :

$$f_y = \frac{(x^3 - 3xy^2)(x^2 + y^2) - (x^3y - xy^3)(2y)}{(x^2 + y^2)^2}.$$

Set $y = 0$:

$$f_y(x, 0) = \frac{(x^3)(x^2) - 0}{(x^2)^2} = \frac{x^5}{x^4} = x.$$

Thus

$$f_{yx}(0, 0) = \lim_{h \rightarrow 0} \frac{f_y(h, 0) - f_y(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h - 0}{h} = 1.$$

Therefore $f_{xy}(0, 0) = -1 \neq 1 = f_{yx}(0, 0)$. □

Problem 3

If $T(x)$ is a linear transformation, find the directional derivative of $T(x)$ at $x = a$ in the direction v .

Solution. For a linear transformation T , we have $T(a + tv) = T(a) + tT(v)$. Then

$$D_v T(a) = \lim_{t \rightarrow 0} \frac{T(a + tv) - T(a)}{t} = \lim_{t \rightarrow 0} \frac{tT(v)}{t} = T(v).$$

Thus the directional derivative is simply $T(v)$. □

Problem 4

If $f(x) = \|x\|^2$ for $x \in \mathbb{R}^n$, find the directional derivative of f at a in the direction v .

Solution. We compute directly:

$$f(a + tv) = \|a + tv\|^2 = (a + tv) \cdot (a + tv) = \|a\|^2 + 2ta \cdot v + t^2\|v\|^2.$$

Hence

$$D_v f(a) = \lim_{t \rightarrow 0} \frac{f(a + tv) - f(a)}{t} = \lim_{t \rightarrow 0} \frac{2ta \cdot v + t^2\|v\|^2}{t} = 2a \cdot v.$$

So $D_v f(a) = 2a \cdot v$. □

Problem 5

$$f(x, y) = \begin{cases} \frac{x^3}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Find the partial derivatives and all directional derivatives of f at $(0, 0)$.

Solution. Partial derivatives at the origin:

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - 0}{h} = \lim_{h \rightarrow 0} \frac{h^3/h^2}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1,$$

$$f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k) - 0}{k} = 0.$$

For a direction $v = (v_1, v_2) \neq (0, 0)$,

$$D_v f(0, 0) = \lim_{t \rightarrow 0} \frac{f(tv_1, tv_2) - 0}{t} = \lim_{t \rightarrow 0} \frac{(tv_1)^3}{(tv_1)^2 + (tv_2)^2} \cdot \frac{1}{t} = \lim_{t \rightarrow 0} \frac{t^3 v_1^3}{t^2(v_1^2 + v_2^2)} \cdot \frac{1}{t} = \frac{v_1^3}{v_1^2 + v_2^2}.$$

Thus every directional derivative exists and equals $\frac{v_1^3}{v_1^2 + v_2^2}$. This is not linear in v , so f is not differentiable at the origin. □

Problem 6

For $f(x, y) = x^2 - y^2$, find the derivative of f at the point $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and hence determine $Df\begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

Solution. The derivative of f is

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (2x, -2y).$$

At $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$,

$$\nabla f(2, 3) = (2 \cdot 2, -2 \cdot 3) = (4, -6).$$

Therefore the derivative as a linear map is

$$Df\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{bmatrix} 4 & -6 \end{bmatrix},$$

which acts on a vector (h, k) by $4h - 6k$. □

Problem 7

For $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $f(x, y) = \begin{pmatrix} xy \\ y^2 \end{pmatrix}$, find $Df\begin{pmatrix} a \\ b \end{pmatrix}$.

Solution. The Jacobian matrix is

$$Df(x, y) = \begin{pmatrix} \frac{\partial}{\partial x}(xy) & \frac{\partial}{\partial y}(xy) \\ \frac{\partial}{\partial x}(y^2) & \frac{\partial}{\partial y}(y^2) \end{pmatrix} = \begin{pmatrix} y & x \\ 0 & 2y \end{pmatrix}.$$

Hence

$$Df\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} b & a \\ 0 & 2b \end{pmatrix}.$$

□

Problem 8

Let $f(x, y) = \begin{pmatrix} xy \\ x^2 - y^2 \end{pmatrix}$. At the point $\mathbf{a} = (1, 1)$, at what rate is f varying in the direction $\mathbf{v} = (1, 0)$?

Solution. The rate of change (directional derivative) of a vector-valued function is given by $Df(\mathbf{a})\mathbf{v}$. Compute the Jacobian:

$$Df(x, y) = \begin{pmatrix} y & x \\ 2x & -2y \end{pmatrix}, \quad \text{so} \quad Df(1, 1) = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix}.$$

Then

$$Df(1, 1)\mathbf{v} = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Thus the vector of directional derivatives of the component functions is $(1, 2)$; i.e., the first component increases at rate 1, the second at rate 2. □

Problem 9

The position of a roller coaster is $(x(t), y(t)) = (\cos t, \sin t)$ and its height is $z = x^2y + x + 1$. Find its rate of change in height with respect to time at time t .

Solution. By the chain rule,

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$

We have

$$\begin{aligned} \frac{\partial z}{\partial x} &= 2xy + 1, & \frac{\partial z}{\partial y} &= x^2, \\ \frac{dx}{dt} &= -\sin t, & \frac{dy}{dt} &= \cos t. \end{aligned}$$

Substituting $x = \cos t$, $y = \sin t$:

$$\frac{dz}{dt} = (2 \cos t \sin t + 1)(-\sin t) + \cos^2 t \cdot \cos t = -2 \cos t \sin^2 t - \sin t + \cos^3 t.$$

□

Problem 10

Given

$$F(t) = \begin{pmatrix} t^2 \\ t^4 \\ -t^3 \end{pmatrix}, \quad G \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \sin(3x_1 + 2x_2 + 5x_3),$$

find $\frac{d}{dt}(G \circ F) \Big|_{t=1}$.

Solution. By the chain rule,

$$\frac{d}{dt}(G \circ F)(t) = \nabla G(F(t)) \cdot F'(t).$$

Compute:

$$\nabla G = (3 \cos(3x_1 + 2x_2 + 5x_3), 2 \cos(3x_1 + 2x_2 + 5x_3), 5 \cos(3x_1 + 2x_2 + 5x_3)).$$

At $t = 1$, $F(1) = (1, 1, -1)$. Then $3x_1 + 2x_2 + 5x_3 = 3(1) + 2(1) + 5(-1) = 0$, so $\cos 0 = 1$. Hence

$$\nabla G(F(1)) = (3, 2, 5).$$

Now $F'(t) = (2t, 4t^3, -3t^2)$, so $F'(1) = (2, 4, -3)$. Thus

$$\frac{d}{dt}(G \circ F) \Big|_{t=1} = (3, 2, 5) \cdot (2, 4, -3) = 6 + 8 - 15 = -1.$$

□

Problem 11

Let

$$F \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u + v \\ u^2 \\ v^3 \end{pmatrix}, \quad G \begin{pmatrix} x \\ y \\ z \end{pmatrix} = xy + z.$$

Find all partial derivatives of $G \circ F$ at $(u, v) = (1, 1)$.

Solution. First compute $H(u, v) = (G \circ F)(u, v) = G(F(u, v))$:

$$H(u, v) = (u + v)(u^2) + v^3 = u^3 + u^2v + v^3.$$

Then

$$\frac{\partial H}{\partial u} = 3u^2 + 2uv, \quad \frac{\partial H}{\partial v} = u^2 + 3v^2.$$

At $(u, v) = (1, 1)$:

$$H_u(1, 1) = 3(1)^2 + 2(1)(1) = 3 + 2 = 5, \quad H_v(1, 1) = 1^2 + 3(1)^2 = 1 + 3 = 4.$$

Thus the partial derivatives are 5 and 4. □

Problem 12

Let

$$F \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u^2 + v \\ uv \end{pmatrix}, \quad G \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ x - y \end{pmatrix}, \quad x = (1, 1)^T.$$

Find $D(G \circ F)|_{x=(1,1)^T}$ and verify that the partial derivatives with respect to u and v of the component functions agree with the entries of this matrix.

Solution. First compute the composition $H = G \circ F$:

$$H(u, v) = G(F(u, v)) = G(u^2 + v, uv) = \begin{pmatrix} (u^2 + v) + uv \\ (u^2 + v) - uv \end{pmatrix} = \begin{pmatrix} u^2 + v + uv \\ u^2 + v - uv \end{pmatrix}.$$

The Jacobian matrix of H is

$$DH(u, v) = \begin{pmatrix} \frac{\partial}{\partial u}(u^2 + v + uv) & \frac{\partial}{\partial v}(u^2 + v + uv) \\ \frac{\partial}{\partial u}(u^2 + v - uv) & \frac{\partial}{\partial v}(u^2 + v - uv) \end{pmatrix} = \begin{pmatrix} 2u + v & 1 + u \\ 2u - v & 1 - u \end{pmatrix}.$$

At $(u, v) = (1, 1)$:

$$DH(1, 1) = \begin{pmatrix} 2(1) + 1 & 1 + 1 \\ 2(1) - 1 & 1 - 1 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 1 & 0 \end{pmatrix}.$$

Now verify using the chain rule: $D(G \circ F)(u, v) = DG(F(u, v)) DF(u, v)$. Compute

$$DG(x, y) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad DF(u, v) = \begin{pmatrix} 2u & 1 \\ v & u \end{pmatrix}.$$

At $(u, v) = (1, 1)$, $F(1, 1) = (2, 1)$. Then

$$DG(2, 1) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad DF(1, 1) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Multiplying:

$$DG(2, 1) DF(1, 1) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 + 1 & 1 + 1 \\ 2 - 1 & 1 - 1 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 1 & 0 \end{pmatrix},$$

which matches $DH(1, 1)$. Hence the partial derivatives agree. □

Problem 13

The temperature is $f(x, y, z) = xy^2 + e^{3xy-z^2}$. A bumblebee moves along a path $g(t)$ with $g(0) = (1, 2, 3)$ and velocity $\mathbf{v} = (-1, 1, 2)$. Find the rate at which the temperature changes at $t = 0$.

Solution. The rate of change of temperature is the directional derivative in the direction of the velocity:

$$\frac{d}{dt}f(g(t))\Big|_{t=0} = \nabla f(g(0)) \cdot \mathbf{v}.$$

Compute the gradient:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = (y^2 + 3ye^{3xy-z^2}, 2xy + 3xe^{3xy-z^2}, -2ze^{3xy-z^2}).$$

At $(1, 2, 3)$:

$$3xy - z^2 = 3 \cdot 1 \cdot 2 - 9 = 6 - 9 = -3, \quad e^{-3} = e^{-3}.$$

Thus

$$\nabla f(1, 2, 3) = (2^2 + 3 \cdot 2e^{-3}, 2 \cdot 1 \cdot 2 + 3 \cdot 1e^{-3}, -2 \cdot 3e^{-3}) = (4 + 6e^{-3}, 4 + 3e^{-3}, -6e^{-3}).$$

Now dot with $\mathbf{v} = (-1, 1, 2)$:

$$\frac{d}{dt}f\Big|_{t=0} = (4 + 6e^{-3})(-1) + (4 + 3e^{-3})(1) + (-6e^{-3})(2) = -4 - 6e^{-3} + 4 + 3e^{-3} - 12e^{-3} = -15e^{-3}.$$

Thus the temperature changes at rate $-15e^{-3}$. □